(1/4)

TRAVEL PLANNER FIRST ITERATION

User Flow

1. users have council meetings
2. an AI note taker apps creates nodes (each nodes represent a potential location/destination along with additional information like budget)
3. compiles it into a story flow chart with different branching story paths
4. users pick a main story path
5. if things go wrong, they can always pick a different "story route"
6. a story flow chart is created for each day
7. a financial plan formulated to show whether its within each users budget

FEATURES

1. Setup

Purpose:

- provide basic information regarding the trip and the people joining the trip.

Collaborators:

- the main user can set up the travel plan and invite their fellow travel mates as collaborators.

Setting up the travel plan:

- setup the basic travel info
- such as the general travel location if it exists (ex. a country or city)
- each user can enter their daily / total budget as well as their home address to find meeting points when the trip starts

2. Discussion Rooms (In-App Meetings)

Purpose:

- act as a single source of truth for all travel plan discussions
- avoid travel plan discussions being split between multiple different group chats and DMs

AI note taker:

- Present during discussion rooms to note down the potential destinations/plans during the discussions.
- Creates **Nodes**
- Nodes formed by the AI will be shown in the chatbox, where users can add additional information.

nodes:

- represent a single destination/location that could be visited during the trip.
- placed into different categories: **Sightseeing, Restaurants, Historical Monuments, Shops, etc.**
- Additional information entered by the users: **Google Maps Link (location), Specified budget for specific location, Additional Pictures (ex. menu prices), user who suggested**
- Nodes can also be created and deleted manually without a meeting

MAIN FEATURE: 3. Decision Tree

Purpose:

- present all potential destinations during the trip in a single decision tree
- Each day of the trip will have its own decision tree

nodes:

- the decision tree will utilized the aforementioned nodes (destinations/locations)
- Nodes will branch out to different nodes, each branching path represents different potential travel plans

arrows:

- connects between nodes
- represents a **travel method** (ex. public transport, car, walking, e-hailing etc.)
- the estimated travel time is represented by the length of the arrow
- each travel method will have an estimated price
- Nodes can have multiple arrows connecting to it, representing different travel methods towards that destination

Main path:

- users can select a main travel route, representing the final plan for the trip
- in the case plans fall through or an unexpected situation occurs, the different branching paths can be seen as backup plans or contingencies

4. Budget Tracker

Purpose:

- based on the currently selected main path and the estimated spending on each travel destination and method
- a full financial plan will be detailed below each day of the decision tree
- Users will be notified when the estimated spending is overbudget for each travel mate

5. Trip Mode

Purpose:

- Serve as a **reliable travel plan** during the actual trip, allowing users to switch routes or plans on the fly.
- Users can **attach images and short notes to each destination** as memories.
- Users can **save receipt pictures and expenses** to track and split bills later.

CHANGES FROM PREVIOUS ITERATION

UPDATED FEATURE 4: **BUDGET TRACKER**

- moved the receipt pictures and expenses (now known as bill uploading) from **5. trip mode** to **4. budget tracker**
- Elaborated on spending notification with coloured visual alerts (red, green, yellow)
- Added individual and group billing system such that users split the estimated bills and view their individual spending more accurately.

UPDATED FEATURE 5: TRIP MODE

- expanded on memories and journalling feature as suggested by mentor session

PROVIDING EXAMPLES

- visual example of what the decision tree **may look like**
- removed **user flow,** as the entire mindmap is one user flow already

(2/4)

TRAVEL PLANNER SECOND ITERATION

1. Setup

Purpose:

- provide basic information regarding the trip and the people joining the trip.

Collaborators:

- the main user can set up the travel plan and invite their fellow travel mates as collaborators.

Setting up the travel plan:

- setup the basic travel info
- such as the general travel location if it exists (ex. a country or city)
- each user can enter their daily / total budget as well as their home address to find meeting points when the trip starts

2. Discussion Rooms (In-App Meetings)

Purpose:

- act as a single source of truth for all travel plan discussions
- avoid travel plan discussions being split between multiple different group chats and DMs

AI note taker:

- Present during discussion rooms to note down the potential destinations/plans during the discussions.
- Creates **Nodes**
- Nodes formed by the AI will be shown in the chatbox, where users can add additional information.

nodes:

- represent a single destination/location that could be visited during the trip.
- placed into different categories:
  - **Sightseeing**
  - **Restaurants**
  - **Historical Monuments**
  - **Shops**
  - **etc.**
- Additional information entered by the users:
  - **Google Maps Link (location)**
  - **Specified budget for specific location**
  - **Additional Pictures (ex. menu prices)**
  - **user who suggested**
- Nodes can also be created and deleted manually without a meeting

MAIN FEATURE: 3. Decision Tree

Purpose:

- present all potential destinations during the trip in a single decision tree
- Each day of the trip will have its own decision tree

arrows:

- connects between nodes
- represents a **travel method** (ex. public transport, car, walking, e-hailing etc.)
- the estimated travel time is represented by the length of the arrow
- each travel method will have an estimated price
- Nodes can have multiple arrows connecting to it, representing different travel methods towards that destination

nodes:

- the decision tree will utilized the aforementioned nodes (destinations/locations)
- Nodes will branch out to different nodes, each branching path represents different potential travel plans

Main path:

- users can select a main travel route, representing the final plan for the trip
- in the case plans fall through or an unexpected situation occurs, the different branching paths can be seen as backup plans or contingencies

4. Budget Tracker

Purpose:

- based on the currently selected main path and the estimated spending on each travel destination and method
- a full financial plan will be detailed below each day of the decision tree
- Users will be notified when the estimated spending is overbudget for each travel mate

Bill manager

- Whoever paid for that bill can upload the bill to the system
- Bill status: individual or sharing
- Addition features is specific adjust for the bills if anyone order different to the rest

Individual and Group Sharing System

Individual

- The bill only deduct in the total budget for that specific users

Group:

- The total amount will be share equally to everyone (unless they did special adjust)
- Auto update the balance

-> At the end of the trip the system will automatically calculate how much the users still owe others people in the group and vice versa

Visual Status Alert

Green: within safe spending limit for all group members

Yellow: warning alert

Red: exceeded spending alert

5. Trip Mode

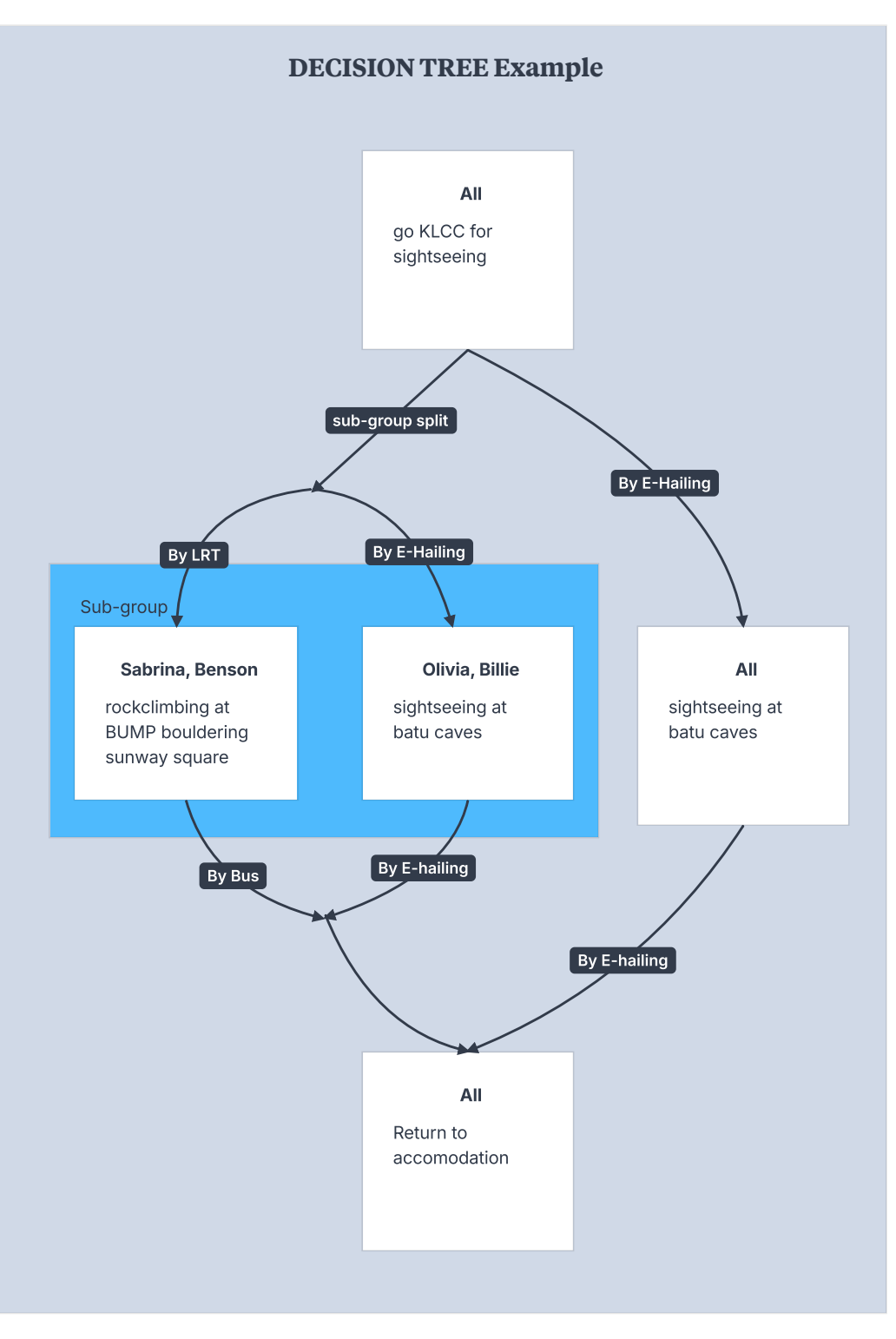
Purpose:

- Serve as a **reliable travel plan** during the actual trip, allowing users to switch routes or plans on the fly.
- Users can **attach images and short notes to each destination** as memories.

Memories & Journalling

- Users can attach photos to completed itinerary planets as “memories”
- Add short descriptions, captions or journal entries for each destination
- Mark activities/destinations as completed
- Users can revisit their memories through the specific itinerary node

DECISION TREE Example



CHANGES FROM PREVIOUS ITERATION

OVERALL **THEMING**:

- from our mentorship sessions, we were suggested to bring over our idea (from our stress manager brain storming even though we ended up picking travel planner) of both **gamifying** the concept (as building a decision tree is already similar to a game)
- not only that, but we decided to pursue a central visual theme for the application as most travel planner apps have a very simple UI.
- Because of our gamification concept, pushing a theme would create more excitement for users
- We settled on a planetary/space concept, where aliens travel on spaceships to new undiscovered planets
- nodes now represent **Planets**
- arrows now represent **Spaceship Paths**
- our AI is the "Spaceships **Autopilot AI** Mode"
- named the application **Galaxseeing**

OTHER CHANGES

1. **Setup**

- rephrased and reorganized this user flow step. Provided more details on what users have to fill in.

2. **Group Planning**

- rephrased and reorganized this user flow step, providing more details.

3. **Decision Tree**

- rephrased and reorganized this user flow step, providing more details.
- separated EDIT and TRIP mode

4. **Expense Tracker**

- rephrased and reorganized this user flow step, providing more details.
- renamed from budget tracker to expense tracker
- as its accessible both when planning the budget and actually during the trip.

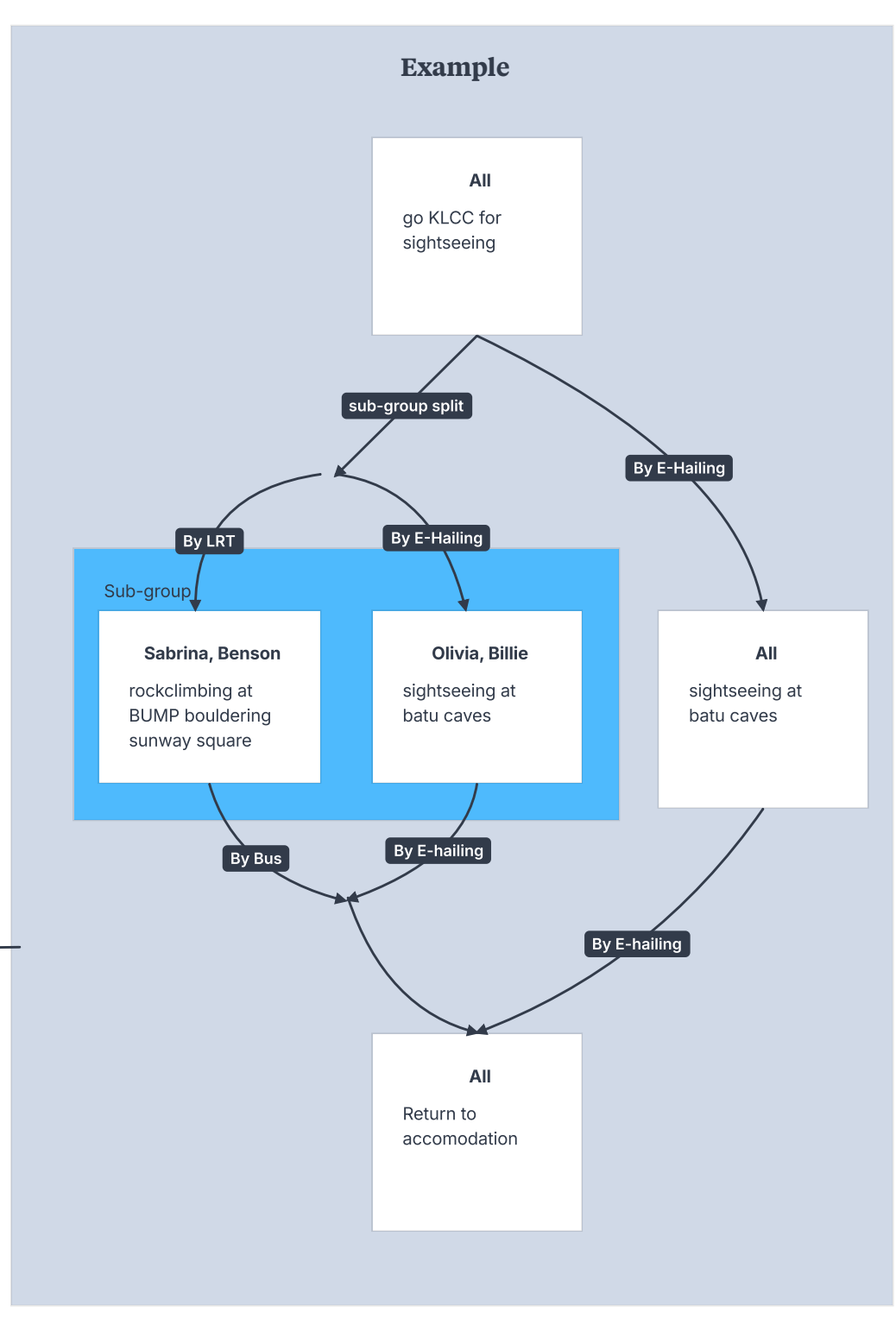
5. **Trip Mode**

- rephrased and reorganized this user flow step, providing more details.
- Added the trip recap feature.
- elaborated on dynamic plan changes where the itinerary reacts in real time to make adjustments and allow users to reroute their path they take in the decision tree.

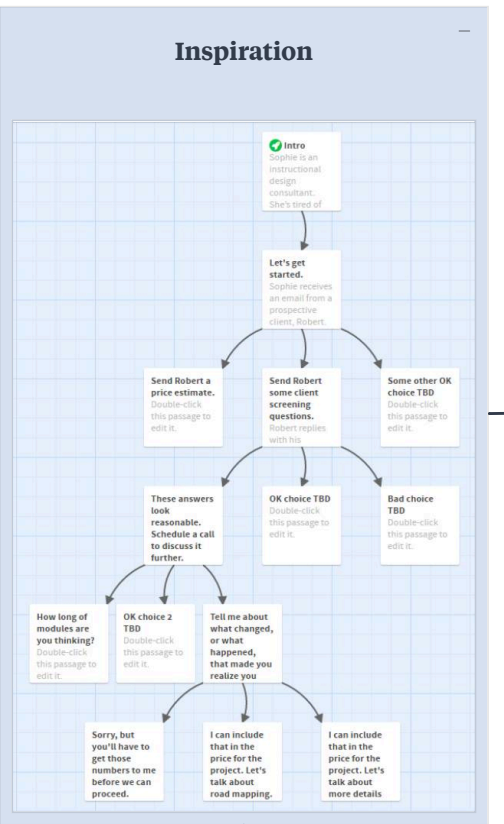
(3/4)

TRAVEL PLANNER THIRD ITERATION

Example



Inspiration



Theming

The app will be built around a **space** theming, where aliens travel across different locations represented by planets.

The app name is **Galaxseeing.**

1. Setup

Create account : User need to create an account and provide basic information

- Name
- Phone number
- Email
- Address
- Date of birth

Create group space: with basic group information

- Number of people
- Trip duration

Invite collaborator

- Search for other users email and send the invitations
- When accept the invitation each user need to provide their budget for for trip

basic trip details:

- General trip destination

For each travel mate:

- daily budget/total budget for the trip
- Home Address
- if they have a personal vehicle that can be used for the trip

2. Group Planning

- Users can hold in-App discussion meetings to discuss the logistics and potential destinations along the trip.

AI planet maker (Autopilot AI):

- Present during discussion rooms to note down the potential destinations/plans during the discussions.
- Creates **Planets**
- Planets formed by the AI will be shown in the chatbox, where users can add additional information

planets:

- represent a single destination/location that could be visited during the trip.

Planets can be placed into different categories:

- **Sightseeing**
- **Restaurants**
- **Historical Monuments**
- **Shops**
- **etc.**

Additional information entered by the users:

- **Google Maps Link (location)**
- **Specified budget for specific location**
- **Additional Pictures (ex. menu prices, potential photo spots)**
- **user who suggested (so that the travel plan can adjust to peoples preferences)**

planets can also be created and deleted manually without a meeting.

3. Decision Tree

- Planets created will be compiled into a flowchart-like Decision Tree for each day of the trip.
- Planets can branch out into other planets to provide several location options.
  - To provide backup plans by following alternative routes when plans fall through.
  - Each possible route is called a **travel path.**
  - The initial Decision Tree will only have branches for **flexible planets** to avoid having too many options, but users can add on more branches for both fixed and flexible planets.
- planets are arranged according to the date, time, distance, category, and priority.

Arrows:

- Represent the potential ***travel method*** (e.g. public transport, car, e-hailing) between two locations, along with the estimated cost and travel time.
- The longer the estimated travel time, the longer the arrow.
- One node can have more than one arrow pointing towards it if there are several ways to reach the location.

Modes:

- ***Edit Mode***: used to select or make changes to planets and routes, mainly used during planning.
- ***Trip Mode***: used during the trip itself, allowing users to view their itinerary easily.

Main Route:

- By selecting planets in Edit Mode, users can form their Main Route, which represents the planned itinerary of the day.
- A financial plan will be shown to see if it accommodates everyone's budgets.

4. Expense Tracker

Divide budget into different categories

the spending amounts is dictated by estimated amounts from online sources

- Accommodation
- Transportation
- Lifestyle
- Food & Beverages

Bill manager

- Whoever paid for that bill can upload the bill to the system
- Bill status: individual or sharing
- Addition features is specific adjust for the bills if anyone order different to the rest

Individual and Group Sharing System

Individual

- The bill only deduct in the total budget for that specific users

Group:

- The total amount will be share equally to everyone (unless they did special adjust)
- Auto update the balance

-> At the end of the trip the system will automatically calculate how much the users still owe others people in the group and vice versa

Visual Status Alert

Green: within safe spending limit for all group members

Yellow: warning alert

Red: exceeded spending alert

5. Trip Mode

Dynamic Plan Changes

- Users can modify their itinerary while the trip is ongoing
- If a planned activity falls through due to bad weather, emergencies, closures, lack of transport, etc., users can find alternative plans
- Use the decision tree to suggest suitable backup activities
- Users can switch to **edit mode** to add, remove, rearrange or replace itinerary planets on the fly

Memories & Journalling

- Users can attach photos to completed itinerary planets as “memories”
- Add short descriptions, captions or journal entries for each destination
- Mark activities/destinations as completed
- Users can revisit their memories through the specific itinerary node

Live Itinerary Adjustment

- Automatically update remaining plans when users are running late or plans change
- Users can drag and rearrange upcoming planets
- Show estimated time remaining for each activity
- Suggest removing or shortening activities if there is not enough time

Trip Recap

- At the end of the trip, compile all completed destinations, photos, journal entries and expenses
- Generate a chronological trip timeline
- Show total spending and spending breakdown
- Create a digital “trip memory” from the collected photos and notes

---

CHANGES FROM PREVIOUS ITERATION

OVERALL **CHANGES**:

- pushed the gamification concept **even further** by involving games in the chatroom and **trip mode,** as suggested by mentors to make bonding before the trip even more of an exciting selling point.
- added a lot of Quality of Life features to accommodate users travel planning and habits, especially in the group planning and decision tree section.

CHANGES PER SECTION

2. Group Planning

- changed from in-app discussion meetings to a chatroom as users are more likely to be available in chatrooms, and is more technically feasible.
- because of such, allows users to paste in external social media links: tiktok, instagram, google maps, etc. for the Autopilot AI to convert those links into travel plans.
- Planets can now be "Fixed" or "Flexible". Fixed Nodes are must-visits (like places with bookings) while flexible are optional.
- Gamification: Added the rocket ship racing game when users are indecisive for the main route of their trip.

3. Decision Tree

- The financial plan feature in the main route is already part of expense tracker.
- added subgrouping feature so users can deviate from their main groups and split up.

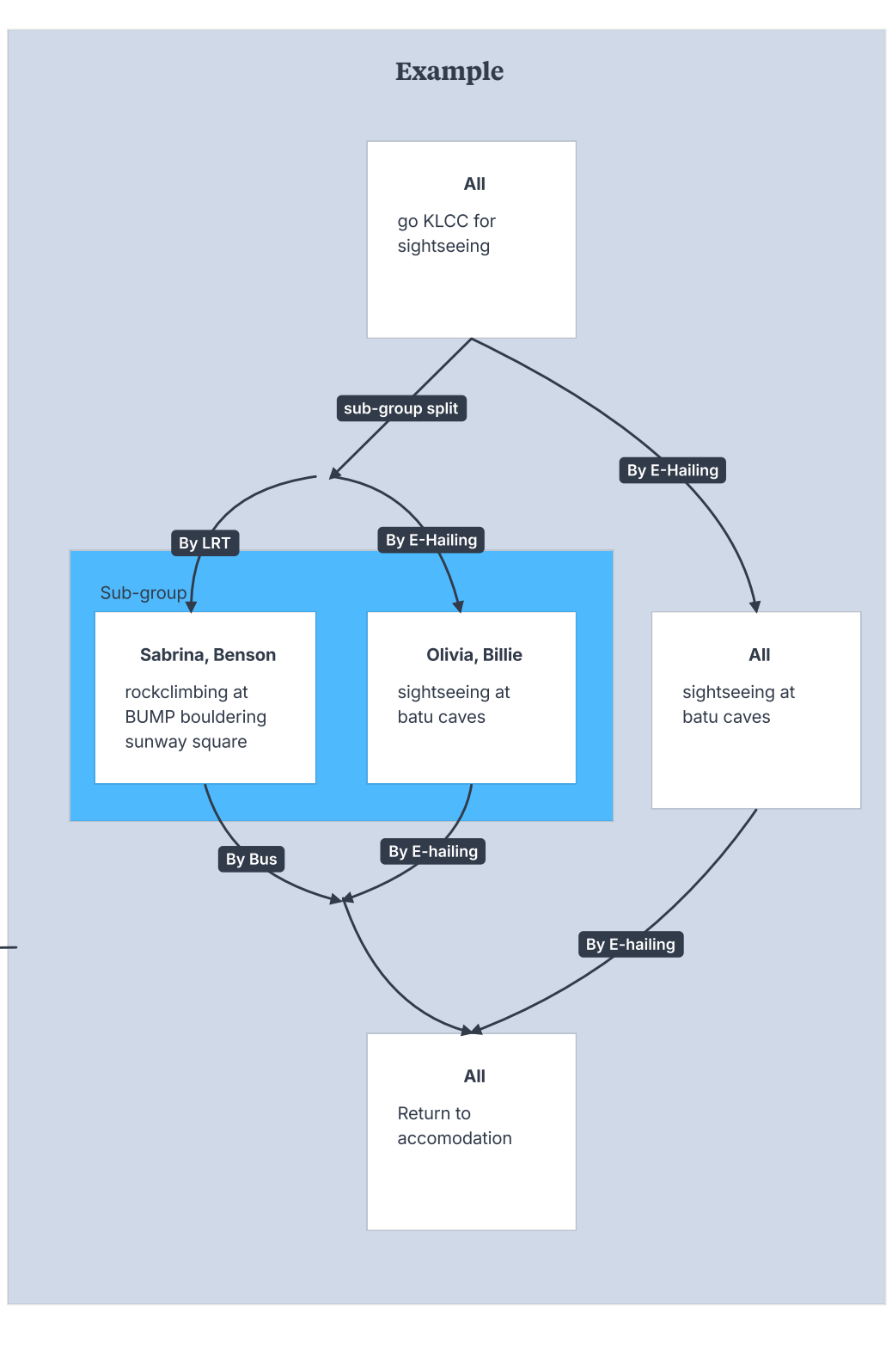
5. Trip Mode

- Live location tracking: users location can be tracked and matched up to the itinerary to mark as visited.
- gamification: when users reach a new location, they are prompted to take as many photos and enter it into the application.

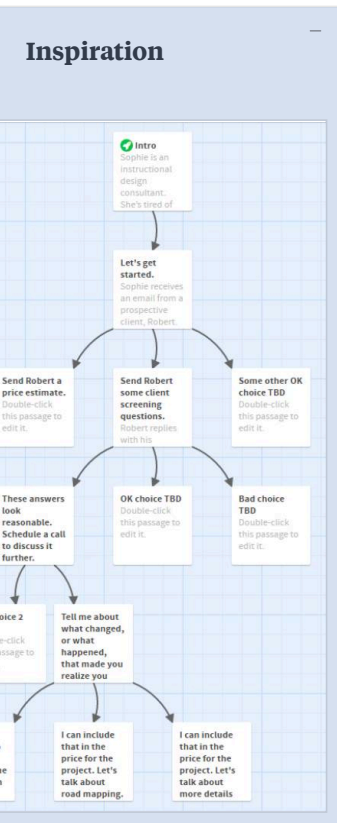
(4/4)

TRAVEL PLANNER FINAL ITERATION

Example



Inspiration



Theming

The app will be built around a **space** theming, where aliens travel across different locations represented by planets.

The app name is **Galaxseeing.**

1. Setup

Create account : User need to create an account and provide basic information

- Name
- Phone number
- Email
- Address
- Date of birth

Create group space: with basic group information

- Number of people
- Trip duration

Invite collaborator

- Search for other users email and send the invitations
- When accept the invitation each user need to provide their budget for for trip

basic trip details:

- General trip destination

For each travel mate:

- daily budget/total budget for the trip
- Home Address
- if they have a personal vehicle that can be used for the trip

2. Group Planning

- Users can hold in-App discussion to discuss the logistics and potential destinations along the trip.
- messages and voice messages
- external links: such as tiktok links and instagram links

AI planet maker (Autopilot AI):

- Present during discussion rooms to note down the potential destinations/plans during the discussions.
- Creates **Planets**
- Planets formed by the AI will be shown in the chatbox, where users can add additional information
- Converts external links such as social media links into **Planets** which are then added to itinerary

planets:

- represent a single destination/location that could be visited during the trip.

Planets can be placed into different categories:

- **Sightseeing**
- **Restaurants**
- **Historical Monuments**
- **Shops**
- **etc.**

Additional information entered by the users:

- **Google Maps Link (location)**
- **Specified budget for specific location**
- **Additional Pictures (ex. menu prices, potential photo spots)**
- **user who suggested (so that the travel plan can adjust to peoples preferences)**

Planets can have a 'fixed' or 'flexible' tag.

- **Fixed:** locations that are a must-visit or have a confirmed appointment.
- **Flexible:** optional locations with alternative planets as a backup.

planets can also be created and deleted manually without a meeting.

Gameification

Additionally, when a node is created by the AI, similar planets based on the budget and user preferences.

If users are unable to decide the planets they want to include, they can play games/space themed minigames to select the planets.

An example would be a space themed "horse racing" app where a random spaceship

3. Decision Tree

- Planets created will be compiled into a flowchart-like Decision Tree for each day of the trip.
- Planets can branch out into other planets to provide several location options.
  - To provide backup plans by following alternative routes when plans fall through.
  - Each possible route is called a **travel path.**
  - The initial Decision Tree will only have branches for **flexible planets** to avoid having too many options, but users can add on more branches for both fixed and flexible planets.
- planets are arranged according to the date, time, distance, category, and priority.

Arrows:

- Represent the potential ***travel method*** (e.g. public transport, car, e-hailing) between two locations, along with the estimated cost and travel time.
- The longer the estimated travel time, the longer the arrow.
- One node can have more than one arrow pointing towards it if there are several ways to reach the location.

Modes:

- ***Edit Mode***: used to select or make changes to planets and routes, mainly used during planning.
- ***Trip Mode***: used during the trip itself, allowing users to view their itinerary easily.

Main Route:

- By selecting planets in Edit Mode, users can form their Main Route, which represents the main planned itinerary of the day.
- Additionally, users can of course still deviate from the main route. It is just the "most suggested" plan of the day for users who enjoy stricter itineraries.

Sub-groups:

- Users can go solo or split into multiple smaller groups by creating and joining a sub-group.
- Each sub-group can choose their own route, where their designated colour will be shown above their selected planets in the Decision Tree.
- A simplified example of a subgroup is shown below.

4. Expense Tracker

Divide budget into different categories

the spending amounts is dictated by estimated amounts from online sources

- Accommodation
- Transportation
- Lifestyle
- Food & Beverages

Bill manager

- Whoever paid for that bill can upload the bill to the system
- Bill status: individual or sharing
- Addition features is specific adjust for the bills if anyone order different to the rest

Individual and Group Sharing System

Individual

- The bill only deduct in the total budget for that specific users

Group:

- The total amount will be share equally to everyone (unless they did special adjust)
- Auto update the balance

-> At the end of the trip the system will automatically calculate how much the users still owe others people in the group and vice versa

Visual Status Alert

Green: within safe spending limit for all group members

Yellow: warning alert

Red: exceeded spending alert

5. Trip Mode

Dynamic Plan Changes

- Users can modify their itinerary while the trip is ongoing
- If a planned activity falls through due to bad weather, emergencies, closures, lack of transport, etc., users can find alternative plans
- Use the decision tree to suggest suitable backup activities
- Users can switch to **edit mode** to add, remove, rearrange or replace itinerary planets on the fly

Memories & Journalling

- Users can attach photos to completed itinerary planets as “memories”
- Add short descriptions, captions or journal entries for each destination
- Mark activities/destinations as completed
- Users can revisit their memories through the specific itinerary node

Live Itinerary Adjustment

- Automatically update remaining plans when users are running late or plans change
- Users can drag and rearrange upcoming planets
- Show estimated time remaining for each activity
- Suggest removing or shortening activities if there is not enough time

Trip Recap

- At the end of the trip, compile all completed destinations, photos, journal entries and expenses
- Generate a chronological trip timeline
- Show total spending and spending breakdown
- Create a digital “trip memory” from the collected photos and notes

Gameification

Users can take photos when they reach a new location with live location tracking. They are given a limited timeframe to do so and can filter out which pictures they want to include.